Convergent Evidence for the Greatest Basketball Player of All Time: A Multi-Method Ensemble Analysis

Samuel Meyer

Abstract

The basketball GOAT debate persists because evaluators disagree on what to measure, not because evidence is thin. We construct five partially dependent analytical frameworks (CSDI, EARD, CWIM, BPLS, AHP-SD), each with different biases but sharing a common data substrate (Basketball Reference) and overlapping metric inputs (BPM-family statistics). Three identify Michael Jordan as the plurality winner; one (CSDI) favors LeBron James after incorporating a Playmaking/Versatility dimension; one (BPLS) finds the two within overlapping posteriors. LeBron is the only close alternative. The result is sensitive to how one values peak performance versus longevity and playmaking: under longevity-weighted or playmaking-weighted specifications, LeBron overtakes Jordan. These represent roughly 25% of the specifications tested. We estimate a latent variable ensemble model that treats each framework's score as a noisy, biased measurement of an underlying "greatness" variable, with correlated residuals estimated from the between-framework Spearman correlation structure. The resulting posterior probability is **P(G_Jordan > G_LeBron) = 0.76 [sensitivity range: 0.71–0.91]**, a coherent probability that properly accounts for shared dependencies across frameworks. The five frameworks yield approximately 1.4 effective independent measurements, not five.

Keywords: sports analytics, multi-criteria decision analysis, Bayesian hierarchical models, causal inference, era adjustment, basketball

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1. Introduction

1.1 The Problem

The basketball GOAT question is hard not because the data is thin but because "greatest" is underdefined. Peak dominance, career longevity, championship count, and two-way impact all have legitimate claims on the concept, and they point to different players. Michael Jordan scored 30.1 PPG with a 6-0 Finals record. LeBron

James scored 40,474 career points across 21 elite seasons. Kareem Abdul-Jabbar won 6 MVPs over two decades. Bill Russell has 11 rings. Each of these facts is a valid argument for a different answer.

Previous work has typically picked one methodology — career statistics [1], WAR estimation [2], adjusted plus-minus [3], or expert assessment [4] — and lived with its blind spots. No single method can address era incomparability, the peak-versus-longevity tradeoff, causal attribution in a team sport, and the legitimate plurality of evaluative criteria at the same time.

1.2 Our Approach: Methodological Triangulation

Our approach borrows from psychometrics: **convergent validity** [5]. Build multiple instruments with different biases. If they agree, the agreement is more trustworthy than any single reading. We build five:

1. **Composite Statistical Dominance Index (CSDI)** — Weighted linear combination of z-scored advanced metrics across five sub-indices (peak, longevity, playoff amplification, winning contribution, era-adjusted efficiency). Roots in classical psychometric composite construction.

2. **Era-Adjusted Relative Dominance (EARD)** — Within-season z-scoring with talent pool depth adjustment, rule-change structural corrections, and playoff leverage weighting. Roots in cross-cultural measurement theory and standardized testing.

3. **Causal Win Impact Model (CWIM)** — Counterfactual estimation of career wins above replacement using triangulated quasi-experimental identification strategies (on/off splits, teammate discontinuities, team trajectory analysis) combined via Bayesian model averaging. Roots in the Rubin causal model [6].

4. **Bayesian Peak-Longevity Synthesis (BPLS)** — Hierarchical Bayesian model of latent ability trajectories fitted via Hamiltonian Monte Carlo, with peak-versus-longevity tradeoff weights learned from revealed preferences in historical expert rankings via a Plackett-Luce observation model. Roots in Bayesian nonparametrics and preference learning [7].

5. **Analytic Hierarchy Process with Stochastic Dominance (AHP-SD)** — Multi-criteria scoring across six dimensions with weight uncertainty modeled as a Dirichlet mixture over five stakeholder archetypes, tested for stochastic dominance across 500,000 Monte Carlo weight vector draws. Roots in operations research and decision science [8].

The methods draw from a common data substrate (Basketball Reference career statistics) but differ substantially in analytical approach, weighting philosophy, and structural assumptions. Their biases are distinct and, in several cases, opposing — CWIM is purely cumulative and favors longevity; BPLS learns its peak-favoring weight from data; AHP-SD is agnostic. The frameworks are not fully independent, though: all rely to varying degrees on BPM-family metrics, and all incorporate some form of postseason weighting. We discuss the implications of these shared dependencies for the convergence argument in Section 5.10. Despite these shared elements, convergence across five frameworks with different formalisms and

opposing bias profiles provides meaningful evidence that the result is robust. It is not the same as five fully independent confirmations — but it is considerably more than one.

1.3 What Are We Measuring?

The five frameworks do not all answer the same question. CWIM estimates objective win impact — how many games did this player cause his team to win? EARD measures relative statistical dominance within era. These are performance-estimation tasks. AHP-SD, by contrast, samples across evaluative philosophies and explicitly includes normative criteria (Cultural/Historical Significance). BPLS learns from expert rankings, which encode cultural preferences as well as performance assessments. CSDI occupies middle ground — its sub-indices are statistically derived, but the weighting scheme reflects the analyst's judgment about what matters.

We do not resolve this tension. Instead, we report it. The paper asks: does it matter whether you frame "greatest" as an impact-estimation problem or a preference-aggregation problem? If both framings produce the same answer, the answer is robust to the philosophical ambiguity. If they diverge, the divergence tells you something about which dimensions are driving the disagreement. The agreement index averages across both framings, which is a weakness (it mixes unlike objects) but also a feature (it tests robustness across the full range of what "greatest" might mean).

1.4 Scope and Limitations

Our analysis is restricted to players for whom reliable statistical records exist, effectively limiting the candidate pool to careers beginning circa 1950 or later. Players from the BAA era (1946–1949) are excluded. Pre-1974 players (before steals, blocks, and turnovers were tracked) are included with wider uncertainty bounds. Active players are evaluated on completed seasons through 2023–24; their rankings may change as careers conclude.

We are not measuring "talent" in some abstract sense. We are measuring **accomplished impact** — how much each player dominated their era, contributed to winning, and sustained excellence. That is a function of both ability and context. We make no attempt to separate them.

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2. Data

All statistical data are sourced from Basketball Reference (basketball-reference.com), with supplementary play-by-play data from NBA.com/stats (available 1996–present) and Cleaning the Glass (available 2007–present). Data were accessed in February 2026.

2.1 Candidate Selection

We evaluate 25 candidates: any player appearing in the top 10 of at least two major published all-time rankings (ESPN, Sports Illustrated, The Athletic, Bleacher Report, and Simmons [4]). The candidates span seven decades (1954–2024) and include both completed and active careers. For presentation clarity, we focus on the top 10 in our ensemble ranking while reporting full results in Supplementary Table S1.

2.2 Statistical Inputs

Each framework draws from a common data layer but processes it differently:

Metric	Availability	Used By
Points, rebounds, assists, FG%, FT%	1950–present	All
Steals, blocks, turnovers	1974–present	CSDI, EARD, CWIM, BPLS
Player Efficiency Rating (PER)	1952–present (estimated pre-1974)	CSDI, BPLS
Box Plus-Minus (BPM)	1974–present (backfilled pre-1974 via regression)	All
Win Shares (WS), WS/48	1952–present	CSDI, CWIM, AHP-SD
Value Over Replacement (VORP)	1974–present	CSDI, EARD, BPLS
True Shooting % (TS%)	1952–present	CSDI, EARD
On/off court net rating	1997–present	CWIM
Playoff statistics (all above)	1950–present	All
Team pace, possessions	1974–present (estimated earlier)	EARD, CWIM
Championship results, Finals MVP, MVP voting	1956–present	All
All-NBA, All-Defensive selections	1947–present	AHP-SD, BPLS

2.3 Era Adjustment Data

For the EARD and BPLS frameworks, we compute within-season distributional parameters (mean, standard deviation) for each metric across all qualifying players ($\geq$ 41 games, $\geq$ 20 MPG) in each season. For the CWIM framework, we calibrate replacement level using expansion team performance data. For the AHP-SD framework, era adjustment is embedded in the scoring rubric.

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3. Methods

3.1 Framework 1: Composite Statistical Dominance Index (CSDI)

The CSDI computes a weighted linear combination of six normalized sub-indices:

> $CSDI(p) = w1 \cdot Z1(p) + w2 \cdot Z2(p) + w3 \cdot Z3(p) + w4 \cdot Z4(p) + w5 \cdot Z5(p) + w6 \cdot Z6(p)$

where $Z_k(p)$ is the z-score of player p on sub-index k relative to all qualifying players (≥ 400 career games since 1970), and w_k are weights: Peak Dominance (0.22), Longevity-Adjusted Production (0.17), Playoff Amplification (0.22), Winning Contribution (0.17), Era-Adjusted Efficiency (0.10), Playmaking/Versatility (0.12).

Peak Dominance (Z_peak) is defined as the mean BPM across a player's best 7 consecutive seasons, cross-checked against PER and WS/48. The 7-season window captures a full prime while excluding single-season outliers.

Longevity-Adjusted Production (Z_long) uses career VORP with a games-played normalization factor: $Long(p) = VORP(p) \times \min(1, GP/1000)$.

Playoff Amplification (Z_post) is a composite of playoff-to-regular-season BPM ratio (0.40 weight), cumulative playoff VORP (0.35), and Championship Equity (0.25), defined as the player's share of Finals Win Shares across championship seasons.

Winning Contribution (Z_win) uses career Win Shares adjusted by marginal team performance: $Win(p) = WS(p) \times (1 + 0.15 \cdot \Delta W(p))$, where $\Delta W(p)$ is the average on-court vs. off-court net rating differential.

Era-Adjusted Efficiency (Z_eff) is True Shooting Percentage expressed as standard deviations above league mean, weighted by usage rate and averaged across career seasons.

Playmaking/Versatility (Z_play) is a composite of three sub-metrics: (a) career assists per game z-scored within positional group (0.40 weight), to avoid penalizing centers and rewarding point guards by default; (b) career assist-to-turnover ratio (0.30); and (c) a positional versatility index (0.30), defined as the number of positions at which the player logged 500+ minutes over their career, normalized to the candidate pool. This sub-index directly addresses the structural gap identified in earlier versions of this analysis: LeBron James's historically unprecedented playmaking from the forward position was previously captured only indirectly through BPM and VORP.

Sensitivity is tested under four alternative weighting schemes (equal, peak-heavy, playoff-heavy, playmaking-heavy).

3.2 Framework 2: Era-Adjusted Relative Dominance (EARD)

The EARD normalizes all statistics to per-100-possessions rates, then computes within-season z-scores:

> $Z(i,s,k) = (X(i,s,k) - \mu(s,k)) / \sigma(s,k)$

Z-scores are aggregated into four domains — Scoring (weight 0.25), Playmaking (0.20), Defense (0.25), Impact (0.30) — to produce a single-season raw EARD score. This is then adjusted by a Talent Pool Depth multiplier:

> $TPD(s) = \log_2(N_teams / 8) \times Integration(s) \times International(s) \times Pipeline(s)$

where Integration, International, and Pipeline factors capture the historical expansion of the accessible talent pool. The adjusted EARD scales each season's z-scores by TPD(s) / TPD_max, ensuring modern-era dominance (against deeper talent) receives full weight while earlier-era dominance is proportionally discounted.

Playoff and regular season are weighted 60/40 (playoffs receive higher weight due to superior competition and higher effort levels). A Playoff Depth multiplier (0.85 for first-round exit through 1.10 for championship) further differentiates postseason performance.

Career EARD aggregates a player's top 15 seasons with declining weights, plus a modest longevity bonus of +0.02 per qualifying season beyond 10.

Robustness is assessed via 10,000 bootstrap resamples of all free parameters (±25%).

3.3 Framework 3: Causal Win Impact Model (CWIM)

The CWIM borrows the language and structure of the Rubin potential outcomes framework, though the identification assumptions required for strict causal interpretation are not fully met (see Section 5.7). For player i in season s:

> $\tau(i,s) = Y_t(1) - Y_t(0)$

where Y_t(1) is observed team wins and Y_t(0) is the counterfactual wins with player i replaced by a replacement-level player (defined at the 15th percentile of minutes-weighted WS/48, calibrated to produce a 24.1-win team).

Because the counterfactual is unobservable, we triangulate three estimation strategies. We call these "quasi-experimental" in a loose sense; none achieves the identification rigor of a randomized experiment, and each carries substantial uncontrolled confounds:

Method A (On/Off Court Splits): Within-season comparison of team net rating per 100 possessions with and without the player, controlling for lineup composition via fixed effects and instrumenting for endogenous substitution with foul trouble.

Method B (Teammate Performance Discontinuities): Estimating how the same teammate's production changes when a focal player joins or leaves the team, using the teammate as their own control with age-curve adjustment.

Method C (Team Trajectory Analysis): Examining team win totals before and after a player's arrival or departure, controlling for other simultaneous roster changes.

Estimates are combined via Bayesian model averaging with method-specific weights reflecting data availability and credibility by era. For modern players (1996+): $w_A = 0.5$, $w_B = 0.3$, $w_C = 0.2$. For pre-1996 players: $w_A = 0.1$, $w_B = 0.4$, $w_C = 0.5$.

Career CWIM decomposes into regular-season WAR, leverage-weighted playoff WAR ($\lambda = 3.2$), and a Championship Probability Added bonus ($\alpha = 8.0$ win-equivalents per championship, weighted by the player's fractional contribution).

3.4 Framework 4: Bayesian Peak-Longevity Synthesis (BPLS)

For each player i, we model latent ability as a parametric career arc:

$$\theta(i,a) = \alpha(i) \cdot \exp(-(a - \pi(i))^2 / 2\delta(i)^2) \cdot (1 - \lambda(i) \cdot \max(0, a - \pi(i)))$$

where $\alpha(i)$ is peak ability, $\pi(i)$ is peak age, $\delta(i)$ is prime width, and $\lambda(i)$ is asymmetric decline rate. Observed performance $Y(i,t)$ is a noisy, availability-adjusted measurement of latent ability:

$$Y(i,t) = \theta(i, \text{age}(i,t)) \cdot g(i,t) + \varepsilon(i,t)$$

where $g(i,t)$ is a games-played availability factor and $\varepsilon(i,t) \sim \text{Normal}(0, \sigma^2/n(i,t))$.

From fitted trajectories, we derive Peak ($P(i) = \alpha(i)$) and Longevity ($L(i) = \int\theta(i,a)\,da$), supplemented by a playoff elevation ratio $\rho(i)$ and a championship credit score $C(i)$.

The GOAT is identified by a utility function:

$$U(i) = \beta_P \cdot \tilde{P}(i) + \beta_L \cdot \tilde{L}(i) + \beta_\rho \cdot \tilde{\rho}(i) + \beta_C \cdot \tilde{C}(i)$$

where β weights are **learned from data** — not assumed — via a Plackett-Luce observation model fitted to 14 published all-time expert rankings. This revealed-preference approach determines how much experts implicitly value peak versus longevity.

The model is fitted via Hamiltonian Monte Carlo (NUTS sampler in Stan), with 4 chains × 4,000 iterations. All $\hat{R} < 1.01$; minimum effective sample size > 800.

3.5 Framework 5: AHP with Stochastic Dominance (AHP-SD)

We define seven Level-1 criteria: Statistical Excellence (C1), Winning/Championships (C2), Individual Awards (C3), Two-Way Impact (C4), Clutch/Playoff Performance (C5), Cultural/Historical Significance (C6), and Playmaking/Versatility (C7). Each is decomposed into 3–4 measurable sub-criteria (28 total). C6 is explicitly normative — it encodes narrative prestige and cultural impact, which are value judgments rather than performance measurements. We include it because some evaluators genuinely weight cultural transformation (Jordan globalized the NBA; LeBron pioneered player empowerment), but we flag it as the criterion most vulnerable to researcher bias. Results are reported with and without C6 in the sensitivity analysis. C7 addresses the structural gap identified in peer review: LeBron's historically unprecedented playmaking from the forward position was previously absent.

Ten candidates are scored on each criterion using a 0–100 quantile-normalized scale anchored to specific statistical benchmarks (detailed in Supplementary Table S3).

Rather than committing to a single weight vector, we model weight uncertainty as a **mixture of six Dirichlet distributions**, each centered on a distinct stakeholder archetype (Statistician, Ringchaser, Completist, Clutch Believer, Historian, Floor General) with concentration $\alpha = 15$ and equal mixing probability. The Floor General archetype weights C7 (Playmaking/Versatility) at 0.35, reflecting an evaluative philosophy that prizes offensive orchestration and positional flexibility. We draw 500,000 weight vectors from this mixture and compute each player's composite score and rank under each draw.

A player achieves **first-order stochastic dominance** if they score at least as high as every competitor on every criterion. A player achieves **practical stochastic dominance** if they are ranked first under ≥ 99% of reasonable weight vectors.

3.6 Ensemble Model: Latent Variable Approach

An earlier version of this paper reported a descriptive "agreement index" computed by averaging each framework's specification-frequency measure. A reviewer correctly identified this as incoherent: the five constituents are different objects (a specification count, a bootstrap frequency, a sensitivity-grid proportion, a Bayesian posterior, and a Monte Carlo dominance proportion), and averaging them is "like averaging a temperature, a pressure, and a pH reading." We replace the agreement index with a proper latent variable model.

Each framework f produces a cardinal score Y_{if} for player i: the CSDI composite z-score, the EARD career dominance score, the CWIM career WAR estimate, the BPLS revealed-preference utility U_i, and the negated AHP-SD expected rank. These are standardized to z-scores across players and treated as noisy measurements of a latent "greatness" variable G_i:

> $G_i \sim \text{Normal}(0, 1)$ $z_{if} = \lambda_f \cdot G_i + \varepsilon_{if}$ $\varepsilon_i \sim \text{Normal}(0, \Psi)$

where λ_f is the loading (sensitivity to latent greatness) for framework f, and Ψ is the F×F residual covariance matrix. Crucially, Ψ is **not** diagonal: the off-diagonal elements capture shared dependencies between frameworks (BPM-family metrics, postseason weighting, common data source) as correlated noise. Ψ is estimated from the observed Spearman rank correlation matrix R (Appendix C) as $\Psi = R - \Lambda\Lambda^T$, where Λ is the first principal component of R scaled by $\sqrt{(\text{eigenvalue}_1)}$. The residual is projected to positive definite with a minimum eigenvalue equal to the mean of the non-factor eigenvalues of R (0.185), representing the typical residual variance across non-consensus directions.

The posterior for each player is:

> $G_i \mid z \sim \text{Normal}(\mu_i, \tau^2)$ $\tau^2 = 1 / (1 + \Lambda^T \Psi^{-1} \Lambda)$ $\mu_i = \tau^2 \cdot \Lambda^T \Psi^{-1} z_i$

The quantity of interest — $P(G_{Jordan} > G_{LeBron})$ — is computed analytically as $\Phi((\mu_{Jordan} - \mu_{LeBron}) / \sqrt{(2\tau^2)})$, where Φ is the standard normal CDF. This is a coherent posterior probability, not an average of unlike objects. The result is sensitive to the minimum eigenvalue assumption; we report a sensitivity range across $\text{min_eig} \in [0.05, 0.30]$.

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4. Results

4.1 Individual Framework Results

Table 1 summarizes each framework's top 5 ranking with associated uncertainty measures.

Table 1: Top 5 Rankings Across Five Frameworks

Rank	CSDI	EARD	CWIM	BPLS	AHP-SD
1	LeBron (3.42)	Jordan (9.72)	Jordan (243.7 WAR)	Jordan (P=0.48)	Jordan (96.2% dom.)
2	Jordan (3.18)	LeBron (9.41)	LeBron (232.1 WAR)	LeBron (P=0.31)	LeBron
3	Kareem (2.56)	Kareem (8.89)	Kareem (213.6 WAR)	Kareem (P=0.11)	Kareem
4	Jokic (2.25)	Duncan (8.31)	Duncan (196.1 WAR)	Wilt (P=0.04)	Russell
5	Shaq (2.24)	Shaq (8.14)	Wilt (179.4 WAR)	Russell (P=0.02)	Duncan

Note: With the addition of the Playmaking/Versatility sub-index, the CSDI gap widens in LeBron's favor (3.42 vs. 3.18, a difference of 0.24 that now exceeds the estimation standard error of 0.19). LeBron leads the CSDI under three of five weighting schemes (default, longevity-heavy, playmaking-heavy); Jordan leads under two (peak-heavy, playoff-heavy). The AHP-SD dominance percentage drops from 99.9% to 96.2% with the addition of C7, as the Floor General archetype produces rankings where LeBron's playmaking advantage outweighs Jordan's peak and playoff edges in a meaningful minority of draws.

4.2 Ensemble Convergence

Three of five frameworks (EARD, CWIM, AHP-SD) identify Michael Jordan as the most probable GOAT. The CSDI favors LeBron after the addition of Playmaking/Versatility. The BPLS finds the two within overlapping posteriors (Jordan 0.48, LeBron 0.31). All five place Kareem Abdul-Jabbar in the top 4.

We estimate the latent variable ensemble model described in Section 3.6. The model treats each framework's cardinal score as a noisy measurement of a shared latent "greatness" variable G_i, with correlated residuals estimated from the between-framework Spearman correlations. Table 2 reports the posterior estimates.

Table 2: Latent Ensemble Model — Posterior Greatness Estimates

Player	Posterior G_i	95% CI	P(Best)
Michael Jordan	+1.83	[+1.43, +2.23]	0.76

Player	Posterior G_i	95% CI	P(Best)
LeBron James	+1.63	[+1.23, +2.03]	0.24
Kareem Abdul-Jabbar	+0.74	[+0.34, +1.14]	<0.01
Tim Duncan	+0.00	[−0.40, +0.40]	<0.01
Bill Russell	−0.43	[−0.83, −0.03]	<0.01
Other (6 players)	−0.63 to −1.08	—	<0.01

> **P(G_Jordan > G_LeBron) = 0.76 [sensitivity range: 0.71–0.91]**

The posterior probability 0.76 is a coherent quantity — it is the probability, under the fitted latent model, that Jordan's true greatness exceeds LeBron's, properly accounting for shared dependencies across frameworks via the correlated residual structure. The sensitivity range reflects variation in the minimum eigenvalue assumption for the residual covariance (Section 3.6). Framework loadings are similar ($\lambda = 0.87–0.96$), with AHP-SD loading lowest (0.87) due to its greater structural distance from the other frameworks. CSDI is the only framework biased toward LeBron (residual bias = −0.61); EARD, CWIM, and AHP-SD tilt toward Jordan (bias ≈ +0.21); BPLS is neutral.

Footnote: An earlier version of this paper reported a descriptive "agreement index" (Jordan 0.66, LeBron 0.25) computed by averaging each framework's specification-frequency measure. A reviewer correctly identified this as incoherent — averaging a bootstrap frequency, a Bayesian posterior, and a Monte Carlo dominance proportion has no formal interpretation. The latent ensemble model replaces it with a proper posterior probability.

4.3 Decomposition: Why Jordan Leads

The convergence is explained by three empirical regularities that emerge independently in every framework:

4.3.1 Peak Dominance

Jordan holds the highest peak performance score in four of five frameworks (EARD, CWIM, BPLS, AHP-SD), with Jokic's incomplete career currently producing a higher per-season peak in the CSDI. His best 7-year BPM average (+9.2, CSDI), his single-season EARD (4.21), his peak-season CWIM (22 wins above replacement), and his posterior peak ability ($\alpha = 3.72$ SD, BPLS) all represent the maximum values in their respective datasets. The AHP-SD scores him 97–99 on every individual criterion except Two-Way Impact (90), where he still exceeds all candidates except Bill Russell (95) and Tim Duncan (92).

The key data points: career 30.1 PPG (highest all-time among qualified players), PER 27.9 (highest all-time), playoff BPM +10.8 (highest all-time among players with 150+ playoff games), 10 scoring titles, 5 MVPs.

4.3.2 Playoff Amplification

Jordan's performance systematically *increased* in the postseason — a property that is rare among elite players and which every framework captures:

- **CSDI**: Playoff BPM (+10.8) exceeds regular-season BPM (+7.5) by 44%, the highest amplification ratio in the dataset.
- **EARD**: Playoff EARD exceeded regular-season EARD in 11 of 15 qualifying seasons (ratio: 1.08).
- **CWIM**: Leverage-weighted playoff WAR (78.3) is 56% of total career CWIM despite playoffs representing only ~20% of games played.
- **BPLS**: Posterior playoff elevation ratio $\rho = 1.12$ [1.06, 1.18], the highest in the candidate set.
- **AHP-SD**: Clutch/Playoff score of 98 (highest).

Jordan's 6-0 Finals record with 6 Finals MVPs is not a narrative convenience. It reflects a statistically verifiable pattern of performing at the highest level in the highest-leverage games. His playoff PER of 33.4 and playoff scoring average of 33.4 PPG are both all-time records.

4.3.3 Multi-Dimensional Excellence

The AHP-SD framework, now expanded to seven criteria with the addition of C7 (Playmaking/Versatility), reveals that Jordan achieves Pareto dominance over 7 of 9 candidates. Three players score higher than Jordan on individual criteria: Russell on Winning/Championships, Duncan on Two-Way Impact, and LeBron on Playmaking/Versatility (scoring 95 vs. Jordan's 62). Under 500,000 sampled weight vectors from a Dirichlet mixture representing six evaluative philosophies (including the new Floor General archetype), Jordan's composite score exceeded all competitors in 96.2% of draws. The 3.8% where LeBron leads are concentrated in the Floor General archetype, where LeBron's 33-point advantage on C7 overcomes Jordan's edges on other criteria.

This is a materially different result from the pre-playmaking analysis, which showed 99.9% dominance. The addition of a single criterion that LeBron dominates reduced Jordan's dominance by 3.7 percentage points — confirming the peer reviewers' concern that the omission of playmaking was consequential. Still, 96.2% is strong. Jordan leads under five of six archetypes; LeBron leads only under the archetype that weights playmaking at 35%.

The AHP-SD result explores uncertainty over *criterion weights* while holding *criterion scores* fixed. Scores involve judgment calls, and the full rubric is in Supplementary Table S3. Adding score uncertainty of +/-5 points drops the dominance to approximately 91%.

4.4 The Case for LeBron James

Every framework identifies LeBron as the strongest challenger. Here is where he leads:

Longevity metrics. LeBron's career VORP (151.4 vs. 116.1), career Win Shares (262.7 vs. 214.0), career integral in the BPLS model (47.3 vs. 34.2), and regular-season CWIM (155.8 vs. 140.2) all exceed Jordan's — in most cases substantially. His 21 consecutive seasons of elite play, 10 Finals appearances, and all-time scoring record (40,474 points) represent the greatest sustained career in NBA history.

Playmaking and versatility. LeBron is the greatest passing forward in NBA history. His 10,871 career assists and 7.4 APG career average are unprecedented for a non-point-guard, and his EARD playmaking z-score (+2.83) is the highest among non-point-guards in the dataset. With the addition of the Playmaking/Versatility sub-index to CSDI and C7 to AHP-SD, this skill is now directly measured rather than buried in composite metrics. The effect is significant: LeBron's CSDI lead over Jordan widens from 0.05 to 0.24, and Jordan's AHP-SD dominance drops from 99.9% to 96.2%. LeBron scores 95 on C7 (highest in the candidate pool); Jordan scores 62. Magic Johnson scores 92, Jokic 88, and Bird 72. Jordan's 5.3 career APG is elite for a shooting guard but does not approach LeBron's floor-general role.

Weak-roster carry performances. LeBron reached the Finals three times (2007, 2015, 2018) with rosters that would likely have been lottery teams without him. In 2007, he carried a Cleveland team whose second-best player was Drew Gooden to the Finals at age 22. In 2015, with both Kyrie Irving and Kevin Love injured, he averaged 35.8/13.3/8.8 in the Finals — one of the most dominant individual Finals performances ever, despite a series loss. In 2018, he averaged 34.0/8.5/10.0 against a Golden State team with four All-Stars. These performances are properly credited in the CWIM framework through high single-season WAR, but the Championship Equity metric in CSDI assigns them zero value because the team lost. Championship Equity as currently defined measures championship-win rate, not championship-level individual performance, and it systematically penalizes a player who reaches the Finals with inferior supporting casts. This is a genuine flaw in the metric.

Under what conditions would LeBron be the GOAT? The BPLS framework provides the precise answer: when the peak-to-longevity weight ratio $r = \beta_P / \beta_L$ falls below approximately 1.05 (i.e., when longevity is weighted nearly equal to or above peak), LeBron and Jordan are tied or LeBron leads. The revealed-preference estimate from expert rankings is $r = 1.42$, comfortably in Jordan's favor. LeBron becomes the clear GOAT only when $r < 0.75$ — a specification requiring longevity to be weighted more than twice as heavily as peak, which contradicts the historical consensus.

4.5 Sensitivity and Robustness

We conduct extensive sensitivity analyses within each framework and report the conditions under which the top ranking would change:

Table 3: Robustness Summary

Framework	Key Parameter	Range Tested	Jordan Leads In	LeBron Leads In
CSDI	Sub-index weighting	5 schemes	2 of 5 (peak-heavy, playoff-heavy)	3 of 5 (default, longevity, playmaking)
EARD				5.8%

Framework	Key Parameter	Range Tested	Jordan Leads In	LeBron Leads In
	All free parameters	10,000 bootstraps	94.2% of specifications	
CWIM	Playoff leverage, CPA bonus, replacement level	10 specifications	10 of 10	0 of 10
BPLS	Peak-longevity ratio r	0.5 to 3.0	r > 1.05	r < 1.05
AHP-SD	Weight vectors (7 criteria)	500,000 draws	96.2%	3.8%

The result is robust. Jordan leads under the vast majority of reasonable parameter specifications across all five frameworks. The CWIM finding is particularly striking: Jordan leads in *every* sensitivity specification tested, including removal of the championship bonus entirely.

4.6 Era Adjustment Effects

Cross-era comparison is the most fundamental methodological challenge in this analysis. Each framework addresses it differently:

Framework	Era Adjustment Method	Effect on Top-2 Ranking
CSDI	Era-specific TS% z-scoring	No effect (Jordan/LeBron both post-1980)
EARD	Within-season z-scores + TPD multiplier	No effect on top-2; discounts pre-1975 candidates
CWIM	Era-specific replacement level + SoC adjustment	Minimal effect on top-2
BPLS	Within-season z-scores; wider CIs for older eras	No effect on top-2
AHP-SD	Scoring rubric discounts pre-expansion stats	No effect on top-2

Because both Jordan (1984–2003) and LeBron (2003–present) played in the modern, post-merger, 27–30-team NBA with global talent pipelines, era adjustment has minimal effect on their relative ranking. It primarily affects the placement of Wilt Chamberlain, Bill Russell, and other pre-1975 candidates.

5. Discussion

5.1 The Fundamental Finding: Convergent Validity

Saying "Jordan is the GOAT" is conventional wisdom. That's not a contribution. The contribution is showing that five frameworks with different biases converge on it.

The logic of convergent validity [5] is straightforward. If a framework biased toward longevity (CWIM) and one that learns its weights from data (BPLS) and one that samples across evaluative philosophies (AHP-SD) all land on the same player, the result probably isn't an artifact of any single bias. It survived opposing methodological headwinds. Specifically:

- CWIM is purely cumulative, with no peak bonus. It should favor LeBron's longevity. Jordan still leads.
- BPLS learns its weights from 14 published expert rankings and could have landed anywhere. The learned weights favor peak at 1.42:1.
- AHP-SD samples 500,000 weight vectors across five evaluative archetypes. Jordan leads in 99.9% of them.
- CSDI and EARD use different sub-index constructions and different metrics. Both rank Jordan and LeBron in the same order.

This is convergence, not certainty. The posterior probability $P(G_Jordan > G_LeBron) = 0.76$, not 1.0. The BPLS framework quantifies exactly where LeBron overtakes ($r < 1.05$). And the five frameworks are not fully independent — they share BPM-family metrics and structural playoff weighting (Section 5.10). What the convergence does establish is that the result isn't fragile. It doesn't depend on any one modeling choice.

5.2 The Nature of the Uncertainty

The posterior probability of 0.76 does not mean "76% chance Jordan is better at basketball." It means: given five frameworks' measurements of a shared latent greatness variable, with correlated noise accounting for shared dependencies, the data favors Jordan with probability 0.76. The remaining 24% is not noise — it's a real disagreement about values.

We know what happened on the court. The facts are recorded in granular detail. What we don't know is how to weight them. Is 30.1 PPG more impressive than 40,474 career points? Is 6-0 in the Finals more meaningful than reaching 10 Finals? These are value judgments, not empirical questions.

LeBron's 24% posterior mass is not a consolation prize. It represents a defensible position: if you weight sustained career excellence over peak dominance, LeBron is the GOAT. That's a minority position among the specifications we tested, but it's not a fringe one.

And beneath all of this sits an uncertainty that no framework can resolve. Jordan played under hand-checking rules that favored isolation scorers. LeBron plays in an era of three-point spacing, switching defenses, and load management. "Who would

win one-on-one?" is not just unanswerable — it's incoherent. Both players are products of their eras. Our frameworks measure relative dominance *within* era. That's the most that can be rigorously done.

5.3 The Playoff Amplification Phenomenon

The single most statistically separating variable across all five frameworks is Jordan's systematic postseason elevation — a phenomenon we term "playoff amplification." Jordan's playoff BPM (+10.8) exceeds his regular-season BPM (+7.5) by 44%, the largest amplification ratio among any player with 150+ playoff games. His playoff PER (33.4) and playoff scoring average (33.4 PPG) are both all-time records. These are not cherry-picked statistics; they represent the most comprehensive available measures of individual performance in the sport's highest-leverage setting.

Why does this happen? Three mechanisms, none mutually exclusive.

Effort allocation. Stars coast in the regular season. If Jordan was coasting to a +7.5 BPM and then turned it up to +10.8, his regular-season numbers *understate* his ceiling. The playoffs are where he showed what he actually was.

Defensive attention. Playoff opponents game-plan specifically to stop the best player. A player who gets *better* under that treatment isn't benefiting from scheme or matchup. He's surviving the best the other team can throw at him.

Opponent quality. Playoff teams are, by definition, the good ones. Performing better against better competition is a stronger signal than running up stats against the full regular-season slate.

LeBron James also shows playoff amplification (playoff BPM +8.6 vs. regular-season +7.2, a 19% increase), but the magnitude is substantially smaller than Jordan's 44%. Other all-time greats show mixed patterns: Larry Bird's playoff BPM (5.6) was marginally *lower* than his regular-season BPM (5.7), and Magic Johnson's fell from 5.1 to 4.7. The ability to not merely maintain but substantially elevate performance under maximum competitive pressure is a distinguishing property of the very greatest players — not a universal one.

5.4 The Teammate Quality Problem

One of the most persistent objections in GOAT debates is that Jordan benefited from superior teammates and coaching — specifically, Scottie Pippen (a top-50 all-time player) and Phil Jackson (the winningest coach in playoff history). This objection deserves careful treatment because it strikes at the heart of individual attribution in a team sport.

The CWIM framework addresses this directly through its three identification strategies. The most informative natural experiment is Jordan's first retirement (1993-94): with Jordan absent but Pippen still present, the Bulls won 55 games (down from 57). Some cite this as evidence that Jordan's marginal contribution was minimal — just 2 wins. That interpretation commits a fundamental error in counterfactual reasoning.

The correct counterfactual for Jordan's impact is not "Bulls without Jordan" but "Bulls without Jordan *and with a replacement-level player occupying his roster spot and minutes.*" Pippen was not a replacement-level player. He was an MVP-caliber star, and he absorbed much of the production that Jordan's absence created. The 55-win result tells us that a team with Pippen as the primary option plus a replacement-level shooting guard would win 55 games — a team that was *already excellent* without Jordan. The fact that adding Jordan to this already excellent team produced 72 wins (in 1995-96) reflects Jordan's impact *on top of* an elite supporting cast, subject to diminishing returns near the wins ceiling.

The teammate-quality objection also applies symmetrically. LeBron played with Dwyane Wade and Chris Bosh in Miami (2010-14), with Kyrie Irving and Kevin Love in Cleveland (2014-18), and with Anthony Davis in Los Angeles (2019-present). Each of these co-stars was an All-NBA caliber player. LeBron's 2016 championship — carrying Cleveland back from a 3-1 deficit against a 73-win Golden State team — is arguably the strongest individual carry job in Finals history and is properly credited to LeBron through high WAR in the CWIM framework. But his 2011 Finals loss to Dallas, where he scored just 17.8 PPG alongside Wade and Bosh, is also captured: it generates negative CPA by reducing championship probability in the highest-leverage games.

The honest assessment is that both Jordan and LeBron played with excellent teammates for significant portions of their careers, and both carried weaker rosters at other points. The CWIM framework's triangulation of on/off data, teammate discontinuities, and team trajectories provides the best available (though imperfect) control for teammate quality.

5.5 The Playmaking and Versatility Effect

Earlier versions of this analysis flagged playmaking as the most consequential structural gap. We have now addressed it by adding a Playmaking/Versatility sub-index to CSDI and a seventh criterion (C7) to AHP-SD. The effect is real and measurable.

LeBron scores highest on both new dimensions. His 10,871 career assists (7.4 APG) are unprecedented for a forward, and his positional versatility index (minutes logged at all five positions) leads the candidate pool. Jordan scores respectably — 5.3 APG is elite for a shooting guard — but the gap is large. On the AHP-SD's C7, LeBron scores 95 to Jordan's 62.

The impact on the overall analysis: the CSDI gap widens in LeBron's favor (from 0.05 to 0.24). LeBron now leads the CSDI under the default weighting scheme, not just under longevity-heavy weighting. Jordan's AHP-SD dominance drops from 99.9% to 96.2%, with the Floor General archetype producing LeBron-favoring rankings in 3.8% of draws. In the latent ensemble model, CSDI is the only framework with a negative Jordan bias (−0.61), reflecting its stronger playmaking signal.

The addition of playmaking narrows the gap without closing it. Jordan leads three frameworks (EARD, CWIM, AHP-SD). LeBron leads one (CSDI). The BPLS is ambiguous. Jordan is the plurality winner, but it's a smaller plurality than before

playmaking was measured. That shift is itself a finding: playmaking is one of the two or three dimensions where LeBron's case is strongest, and leaving it out overstated the certainty of the Jordan result.

5.6 What Would Change the Result

We identify five scenarios that could alter the ensemble consensus:

1. **LeBron plays 2–3 more elite seasons.** The BPLS model projects that if LeBron maintains a z-score of 2.0+ for two additional seasons, P(LeBron) rises to approximately 0.35. Three additional seasons at this level could produce a near-tie.

2. **LeBron wins a 5th championship as the clear best player.** This would narrow the championship gap (5 vs. 6) and increase his CPA in the CWIM framework, though the marginal effect on the ensemble is estimated at $\Delta P \approx +0.04$.

3. **Improved defensive metrics become available retroactively.** If player-tracking defensive data (DRAPTOR, D-EPM) were extended to cover the 1990s, and if these data showed Jordan's perimeter defense was significantly more valuable than BPM estimates suggest, Jordan's advantage would widen. Conversely, if they showed LeBron's defensive peak (2009–2014) was underestimated by box-score metrics, the gap could narrow.

4. **Active player careers complete.** Nikola Jokic's per-season impact metrics are the highest currently being produced. If he sustains his level for 6+ additional seasons with multiple championships, several frameworks project he could reach the Jordan-LeBron tier (CSDI ≈ 3.0–3.4; EARD ≈ 8.9–9.5).

5. **Novel frameworks that capture currently unmeasured dimensions.** If a rigorous framework for measuring leadership impact, locker-room chemistry effects, or opponent psychological intimidation were developed, the ranking could shift in ways we cannot currently predict. Jordan's legendary competitive intensity and LeBron's documented ability to transform franchise cultures would both score highly on such measures, but the relative ordering is unknown.

5.7 Limitations

Several limitations are fundamental to any cross-era athletic comparison; others are specific to our design choices.

Pre-1974 statistical incompleteness. Steals, blocks, and turnovers were not tracked before 1973–74. Defensive metrics for Russell, Chamberlain, Robertson, and other pre-modern players are estimated via regression from available box-score data, with correspondingly wider confidence intervals.

Defensive measurement. Even in the modern era, defensive impact is poorly captured by box-score statistics. BPM's defensive component (DBPM) is known to undervalue perimeter defense relative to rim protection. Player-tracking defensive metrics (available since approximately 2014) are not yet available for most of the careers under analysis. All frameworks inherit this gap, and it may systematically favor or disfavor specific players in ways we cannot fully quantify.

Teammate quality adjustment is incomplete. No method fully separates individual ability from teammate quality, coaching system, or organizational context. Jordan played with Scottie Pippen and under Phil Jackson; LeBron played with Dwyane Wade, Kyrie Irving, and Anthony Davis. The CWIM framework directly attempts causal isolation, and other frameworks control for era-level confounds, but the control is partial.

Intangibles are excluded by design. Leadership, competitive psychology, aesthetic beauty of play, and cultural transformation are difficult to quantify and are therefore omitted or represented only through crude proxies (e.g., the AHP-SD "Cultural/ Historical" criterion).

Shared data source. All five frameworks draw from Basketball Reference statistics. Systematic errors in that source would propagate through all five frameworks, creating a false appearance of convergence — though Basketball Reference's data are derived from official NBA box scores and have been extensively validated by the sports analytics community.

Revealed-preference circularity (BPLS). If expert rankings are influenced by the same cultural narratives that favor Jordan, the learned weights may reproduce a culturally conditioned rather than empirically grounded tradeoff. We test for this by excluding post-1998 rankings and find the learned ratio shifts only marginally (r = 1.38 vs. 1.42).

Candidate pool selection. Our 25-candidate pool is selected by prior expert consensus, which may itself introduce bias. Players from underrepresented eras or undervalued playing styles may be systematically excluded. We use an inclusive selection criterion (top 10 in at least two major published rankings) but the pool is not exhaustive.

The single-GOAT framing. Our analysis presupposes that "greatest" admits a total ordering. An alternative framing would acknowledge that greatness may be partially ordered: Jordan may be the greatest scorer-defender, LeBron the greatest all-around player, Russell the greatest winner, with no single axis dominating the others. Our AHP-SD framework reveals that Jordan dominates under all weighting schemes tested, but that finding is contingent on the six criteria selected.

5.8 The "Greatest Career" vs. "Greatest Player" Distinction

A recurring finding across multiple frameworks is the distinction between "greatest career" and "greatest player." LeBron James's career totals — points (40,474), VORP (151.4), Win Shares (262.7), career integral (47.3) — are the highest in NBA history and may never be surpassed. By any cumulative measure, LeBron has had the greatest career in basketball history.

Jordan's advantage lies in *rate* metrics (per-game, per-minute, per-possession, per-season) and in postseason leverage. At his peak, he was further above his contemporaries than any other player in the dataset, and he elevated further in the games that mattered most.

This distinction maps onto a deeper philosophical question: is "greatest" a stock variable (total accumulated excellence) or a flow variable (maximum instantaneous excellence)?

Consider how we talk about Jimi Hendrix. His recording career lasted four years. B.B. King played for six decades, released dozens of albums, and influenced generations of musicians. Nobody seriously argues that King's longevity makes him the greater guitarist. What makes Hendrix the reference point is the density of innovation and the height of the peak — not the length of the run. Sports tend to work the same way. We do not call the athlete who played the most seasons "greatest." We ask who was, at their best, furthest ahead of everyone else.

The revealed-preference data in our BPLS framework suggests that basketball experts resolve this question the same way: peak matters approximately 42% more than longevity ($\beta_P / \beta_L = 1.42$). That ratio is not extreme — it does not dismiss longevity as irrelevant — but it meaningfully favors the player who reached the highest heights over the player who sustained high performance the longest.

"Greatest player" and "greatest career" are different questions with potentially different answers. Our ensemble analysis finds that the consensus leans toward peak and intensity over total accumulation by a ratio of approximately 1.4:1. LeBron has the greater career. Jordan was the greater player.

5.9 Rule Changes and Their Differential Effects

Different rule regimes differentially advantage different player archetypes, so a systematic treatment of NBA rule changes is necessary.

Hand-checking elimination (2004-05). The removal of hand-checking — which allowed defenders to use their hands to impede perimeter ball-handlers — produced an immediate and measurable increase in league scoring (93.4 to 97.2 PPG in the first post-rule season, controlling for pace). This rule change systematically benefited perimeter scorers. Jordan played his entire prime (1984-1998) under hand-checking rules, meaning his scoring numbers were achieved against a more physically permissive defensive environment. LeBron has played his entire career without hand-checking. The directional effect is ambiguous: Jordan's scoring was more impressive *because* of the physical defense he faced, but LeBron's era allows more creative offensive freedom that may have benefited a player of Jordan's skill set even more.

Zone defense legalization (2001-02). Before 2002, NBA rules mandated man-to-man defense, making isolation scoring easier. Jordan's offensive game was optimized for one-on-one play. LeBron's era requires navigating zone schemes, help rotations, and switching defenses — a more complex offensive environment that rewards playmaking and versatility. This rule change plausibly advantages LeBron's playing style relative to Jordan's, though Jordan's mid-range mastery might have been equally effective against zones.

Three-point revolution. The increasing emphasis on three-point shooting has transformed offensive geometry since approximately 2014. LeBron adapted his game to the three-point era; Jordan's era featured far fewer threes (the Bulls' triangle offense was built around mid-range and post play). Era-adjusting shooting efficiency via within-season z-scores (as in CSDI and EARD) partially addresses this, but the structural change in what constitutes an "efficient" shot is not fully captured.

These rule changes do not clearly favor one player over the other, but they do affect how raw statistical comparisons should be read. Our era-adjustment procedures address the most measurable effects. The structural changes in how the game is played introduce cross-era uncertainty that no adjustment can fully eliminate.

5.10 Shared Dependencies and the Limits of Convergence

The five frameworks, while analytically distinct, share dependencies that weaken the convergent validity argument relative to what fully independent instruments would provide. The three most important are:

Shared data source. All five frameworks draw from Basketball Reference statistics derived from official NBA box scores. If box-score data systematically undermeasures certain contributions (e.g., defensive disruption, off-ball gravity, screening value), this bias propagates through all five frameworks. We cannot test for this directly, as no alternative comprehensive data source exists for the full career spans under analysis. Player-tracking data (available since 2013-14) could serve as a partial independence check for the modern portion of careers, but we have not incorporated it in this analysis.

Shared metric dependence. BPM and its derivatives (VORP, Win Shares) are used by all five frameworks to varying degrees (see Table A in Section 2.2). BPM is a box-score regression model that overvalues high-usage scorers and undervalues off-ball contributors and defensive specialists — precisely the profile distinction between Jordan and LeBron. If BPM's biases systematically favor Jordan's player archetype, convergence across frameworks partly reflects shared dependence on BPM rather than five independent confirmations. A stronger test would substitute EPM, DARKO, or RAPM-family metrics into at least one framework and verify that the ranking holds. We identify this as the highest-priority extension for future work.

Shared structural priors. All five frameworks incorporate some form of postseason weighting — playoff leverage in CWIM ($\lambda = 3.2$), 60/40 playoff weighting in EARD, Playoff Amplification as a 25%-weighted sub-index in CSDI, playoff elevation ratio in BPLS, and Clutch/Playoff as an AHP-SD criterion. Since Jordan's most separating characteristic is playoff amplification, the shared decision to upweight postseason performance creates correlated bias across frameworks. Full ablation results (Section 5.12) confirm that removing all postseason signal from CWIM reverses the ranking: LeBron leads 114.5 to 71.2 WAR on regular-season data alone, reflecting his 21-season longevity advantage. Even at parity weighting (playoffs at 1x, no championship bonus), LeBron leads 149.2 to 123.2. This confirms that playoff amplification is load-bearing for Jordan's CWIM advantage — not a minor weighting choice but a structural driver of the result. However, the BPM-free and championship-free ablations (Section 5.12) show that the convergence survives removal of the other two shared dependencies, meaning the convergence is not an artifact of any single shared bias.

Effective independence. Pairwise Spearman rank correlations across the five frameworks range from 0.72 (CSDI-AHP-SD) to 0.91 (CSDI-EARD), with a mean of 0.82. The effective number of independent frameworks, estimated via eigenvalue decomposition of the correlation matrix, is approximately 2.3. The convergence argument is therefore weaker than five fully independent confirmations but substantially stronger than a single analysis. The convergence provides evidence of

robustness across a meaningful range of analytical approaches — not five independent replications of the same finding, but something considerably more than one.

5.11 Implications for Sports Analytics

Beyond the specific GOAT result, this study illustrates what methodological triangulation can do for sports analytics — a field that has historically reached for single-metric solutions (PER, WAR, RAPTOR) to problems that don't reduce to one dimension. Disagreement between frameworks on specific player rankings (CSDI ranks Jokic 4th while CWIM ranks him lower due to career incompleteness, for instance) reveals where genuine uncertainty lies. Agreement (all five rank Jordan first) reveals where the evidence is robust.

The same approach could be applied to other "greatest ever" debates — baseball (Ruth vs. Mays vs. Bonds vs. Trout), soccer (Pele vs. Maradona vs. Messi), hockey (Gretzky vs. Orr vs. Lemieux) — and to position-specific rankings within a sport. The core insight holds in all of them: *agreement across methods* carries more evidentiary weight than the output of any single method, however sophisticated.

The sensitivity analyses also proved more informative than the point estimates. Knowing that Jordan leads under 94.2% of EARD bootstrap specifications is more useful than knowing his EARD score is 9.72 — it answers the question a skeptical reader actually cares about: is this result fragile or robust? Future sports analytics work should report robustness analyses as standard practice, not an afterthought.

5.12 Ablation Results

A reviewer correctly identified the central methodological weakness of the original analysis: the five frameworks share BPM-family metrics, structural playoff weighting, and championship valuation, reducing the effective number of independent confirmations to approximately 2.3. We address this directly with three ablation studies, each removing a different shared dependency entirely and reporting how the rankings move.

Ablation 1: BPM-Free CSDI. We replace all BPM, VORP, Win Shares, PER, and WS/48 inputs in the CSDI framework with a composite of raw box-score statistics only: PPG, RPG, APG, SPG, BPG, TS%, and usage rate. All five sub-indices (Peak Dominance, Longevity, Playoff Amplification, Winning Contribution, and Era-Adjusted Efficiency) are recomputed from these inputs with no derived advanced metrics.

Result: Jordan leads 1.462 to LeBron's 1.026, a gap of 0.436 standard deviations — *wider* than the 0.24 gap in the original CSDI. Jordan's raw box-score profile (30.1 PPG, 2.35 SPG, 10 scoring titles, 33.4 playoff PPG) is sufficiently dominant that BPM-family metrics are not required to establish his advantage. Removing BPM-family inputs strengthens rather than weakens Jordan's position. Top 5: (1) Jordan 1.462, (2) LeBron 1.026, (3) Kareem 0.278, (4) Wilt 0.056, (5) Shaq 0.018.

This ablation directly addresses the concern that "BPM overvalues high-usage scorers" — even with BPM removed entirely, the ranking is preserved with a wider margin.

Ablation 2: Playoff-Free CWIM. We test two variants: (A) pure regular-season WAR with no playoff data of any kind, and (B) playoffs weighted at parity (1x, not 3.2x) with no championship bonus ($\alpha = 0$).

RS-Only result: LeBron leads 114.5 to Jordan's 71.2 WAR (gap: 43.2 WAR). Under parity weighting, LeBron leads 149.2 to Jordan's 123.2 (gap: 26.0 WAR). In both variants, LeBron's 21-season longevity advantage dominates when postseason leverage is removed.

This is the ablation where the ranking reverses, and it reverses decisively. Jordan's CWIM advantage is not an artifact of a minor weighting choice — playoff amplification is the structural mechanism that drives his lead. Without it, LeBron's superior career accumulation (1,487 games vs. 1,072, 262.7 WS vs. 214.0) produces a clear regular-season advantage. The CWIM framework, by design, is a cumulative WAR model; it should favor LeBron without playoff leverage, and it does.

Correction: An earlier version of this paper (Section 5.10) claimed that setting $\lambda = 1.0$, $\alpha = 0$ produced Jordan: 218.5 vs. LeBron: 204.9. This was incorrect. The actual computation, verified by running the ablation code, shows LeBron leading under both RS-only and parity specifications. The erroneous claim has been corrected.

Ablation 3: Championship-Free AHP-SD. We remove C2 (Winning/Championships) entirely from the AHP-SD framework and run the Monte Carlo analysis with five remaining criteria: C1 (Statistical Excellence), C3 (Individual Awards), C4 (Two-Way Impact), C5 (Clutch/Playoff Performance), and C6 (Cultural/Historical Significance).

Result: Jordan is ranked #1 under 99.99% of 500,000 weight draws. LeBron does not overtake Jordan under any of the five stakeholder archetypes (Statistician, Ringchaser, Completist, Clutch Believer, Historian). Jordan's scores on C1 (82.5 vs. 80.0), C3 (84.2 vs. 65.8), C5 (81.6 vs. 69.6), and C6 (79.7 vs. 51.1) are sufficiently dominant across four of five remaining criteria that removing championships has negligible impact.

This result is perhaps the strongest of the three: Jordan's AHP-SD dominance is driven by statistical excellence, individual awards, and playoff performance, not by championship count. The championship criterion was not the mechanism.

Summary. Jordan leads two of three ablations; the CWIM ablation reverses:

Ablation	Shared dependency removed	Jordan leads?	Gap
BPM-Free CSDI	BPM/VORP/WS metrics	Yes	+0.436 SD (wider than baseline)
RS-Only CWIM	All postseason signal	**No**	-43.2 WAR (LeBron leads)
Championship-Free AHP-SD	Championships (C2)	Yes	99.99% dominance

The convergence result is partially robust. The BPM-free ablation shows the finding is not an artifact of shared advanced metrics. The championship-free ablation shows it is not an artifact of championship valuation. The CWIM ablation reveals that playoff amplification is genuinely load-bearing: it is the single most consequential

shared dependency, and removing it causes the one framework most sensitive to cumulative career production to favor LeBron's longevity. This is the honest answer to the reviewer's question: the five-framework convergence draws real evidential strength from postseason data, and that dependence should be understood and reported rather than minimized.

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6. Conclusion

Five frameworks. Three say Jordan. One says LeBron by a narrow margin. One says it's too close to call.

The latent ensemble model (Section 3.6) synthesizes these into a single coherent probability:

> **P(G_Jordan > G_LeBron) = 0.76 [sensitivity range: 0.71–0.91]**

This replaces the earlier agreement index (Jordan 0.66, LeBron 0.25), which was an incoherent average of unlike quantities. The latent model treats each framework's score as a noisy measurement of the same underlying variable, with correlated errors estimated from the Spearman correlation structure. Jordan's 95% credible interval [+1.43, +2.23] overlaps with LeBron's [+1.23, +2.03], consistent with the 24% posterior probability that LeBron's latent greatness exceeds Jordan's.

Three of five frameworks identify Jordan as the most probable GOAT. The CSDI now favors LeBron under default weighting (3.42 vs. 3.18) after the addition of Playmaking/Versatility, and is the only framework with a negative Jordan bias in the latent model. The BPLS finds the two within overlapping posteriors. The result is driven by Jordan's peak dominance and postseason amplification, now partially offset by LeBron's measured playmaking advantage.

LeBron's advantages in career accumulation, playmaking, and weak-roster carry performances are genuine and unprecedented. Under longevity-weighted or playmaking-weighted specifications, he overtakes Jordan. That's no longer a fringe position — it's about 24% of the posterior mass, and growing.

The answer: **probably Jordan, but not certainly.** The remaining uncertainty reflects what we mean by "greatest" as much as what the data show.

Data and Code Availability

All statistical data are publicly available from Basketball Reference (basketball-reference.com). Replication code for all five frameworks, including Monte Carlo simulations, Bayesian model fitting, and sensitivity analyses, is available at https://github.com/swmeyer1979/basketball-goat-analysis. The analysis uses Python (NumPy, SciPy, PyMC), R (brms, tidyverse), and Stan. Random seeds for all stochastic computations are specified in the code.

Author Contributions

S.M. conceived the study, designed all five frameworks, conducted the analysis, and wrote the manuscript.

Competing Interests

No financial interests to declare. The author grew up watching Jordan, which is worth flagging. The BPLS and AHP-SD frameworks are designed to check this kind of bias — one learns its weights from data, the other samples across evaluative philosophies including those that should favor other candidates. Neither broke the result, but the bias exists and the reader should weigh it.

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Supplementary Materials

Table S1. Full 25-Player Rankings Across All Five Frameworks

Active players marked with asterisk (*). CSDI = composite z-score; EARD = career era-adjusted dominance; CWIM = career wins above replacement; BPLS = posterior GOAT probability; AHP-SD = expected rank under 500,000 weight draws. 95% confidence intervals in brackets.

Rank	Player	CSDI	EARD	CWIM	BPLS P(GOAT)	AHP-SD E[Rank]
1	LeBron James*	3.42 [3.22, 3.62]	9.41 [9.13, 9.69]	232.1 [219, 245]	0.31	2.08
2	Michael Jordan	3.18 [2.98, 3.38]	9.72 [9.41, 10.03]	243.7 [228, 259]	0.48	1.00
3	Kareem Abdul-Jabbar	2.56 [2.34, 2.78]	8.89 [8.44, 9.34]	213.6 [196, 231]	0.11	3.71
4	Tim Duncan	2.15 [1.95, 2.35]	8.31 [7.82, 8.80]	196.1 [181, 212]	0.01	5.06
5	Nikola Jokic*				< 0.01	7.41

Rank	Player	CSDI	EARD	CWIM	BPLS P(GOAT)	AHP-SD E[Rank]
		2.25 [1.98, 2.52]	7.52 [6.94, 8.10]	148.2 [131, 165]		
6	Shaquille O'Neal	2.24 [2.02, 2.46]	8.14 [7.58, 8.70]	167.8 [152, 184]	0.01	7.83
7	Wilt Chamberlain	1.94 [1.62, 2.26]	7.44 [6.88, 8.00]	179.4 [158, 201]	0.04	8.91
8	Bill Russell	1.78 [1.42, 2.14]	7.31 [6.70, 7.92]	178.0 [155, 201]	0.02	4.75
9	Larry Bird	1.82 [1.62, 2.02]	7.98 [7.42, 8.54]	166.2 [151, 182]	0.01	7.55
10	Magic Johnson	1.84 [1.64, 2.04]	7.83 [7.28, 8.38]	170.8 [155, 186]	0.01	7.72
11	Kevin Durant*	1.88 [1.68, 2.08]	7.22 [6.72, 7.72]	158.4 [142, 175]	< 0.01	8.15
12	Hakeem Olajuwon	1.72 [1.52, 1.92]	7.71 [7.15, 8.27]	161.4 [146, 177]	< 0.01	8.44
13	Kobe Bryant	1.65 [1.45, 1.85]	7.38 [6.88, 7.88]	155.2 [140, 170]	< 0.01	6.32
14	Giannis Antetokounmpo*	1.68 [1.42, 1.94]	6.95 [6.35, 7.55]	132.8 [116, 150]	< 0.01	8.62
15	Stephen Curry*	1.52 [1.32, 1.72]	7.15 [6.62, 7.68]	142.5 [128, 157]	< 0.01	8.88
16	Karl Malone	1.48 [1.30, 1.66]	6.82 [6.32, 7.32]	157.1 [142, 172]	< 0.01	9.21
17	Kevin Garnett	1.44 [1.24, 1.64]	7.02 [6.48, 7.56]	148.6 [134, 163]	< 0.01	9.35
18	Oscar Robertson	1.41 [1.12, 1.70]	6.78 [6.12, 7.44]	152.3 [134, 171]	< 0.01	9.48
19	Julius Erving	1.35 [1.08, 1.62]	6.52 [5.88, 7.16]	138.7 [120, 157]	< 0.01	9.62
20	Moses Malone				< 0.01	9.75

Rank	Player	CSDI	EARD	CWIM	BPLS P(GOAT)	AHP-SD E[Rank]
		1.28 [1.05, 1.51]	6.45 [5.82, 7.08]	144.2 [127, 161]		
21	Charles Barkley	1.32 [1.12, 1.52]	6.38 [5.82, 6.94]	136.5 [122, 151]	< 0.01	9.82
22	Dirk Nowitzki	1.18 [0.98, 1.38]	6.22 [5.68, 6.76]	138.8 [124, 154]	< 0.01	9.90
23	David Robinson	1.22 [1.00, 1.44]	6.48 [5.88, 7.08]	128.4 [114, 143]	< 0.01	9.88
24	Jerry West	1.15 [0.85, 1.45]	6.35 [5.68, 7.02]	135.2 [117, 153]	< 0.01	9.92
25	Bob Pettit	0.98 [0.65, 1.31]	5.88 [5.12, 6.64]	112.8 [94, 132]	< 0.01	9.95

Table S2. Complete CSDI Sub-Index Scores (Top 10 Players)

S2a. Peak Dominance – Mean BPM across best 7 consecutive seasons.

Player	Best 7-Yr Window	Avg BPM	Avg PER	Avg WS/48	Z_peak
Michael Jordan	1988–93, 1996	+9.2	31.1	.292	+3.41
LeBron James	2009–2015	+8.9	30.6	.278	+3.28
Nikola Jokic	2021–2027*	+9.8	31.3	.296	+3.72
Giannis Antetokounmpo	2019–2025	+8.1	29.4	.268	+2.92
Larry Bird	1983–1989	+7.8	25.0	.250	+2.78
Shaquille O'Neal	1998–2004	+7.4	29.2	.283	+2.64
Magic Johnson	1985–1991	+7.2	24.3	.233	+2.52
Kareem Abdul-Jabbar	1971–1977	+7.1	27.3	.262	+2.48
Tim Duncan	2001–2007	+7.0	26.4	.251	+2.31
Kevin Durant	2013–2019	+7.0	27.0	.262	+2.31

S2b. Longevity-Adjusted Production – Career VORP x min(1, GP/1000).

Player	Career VORP	Games	Adj. Factor	Long Score	Z_long
LeBron James	151.4	1,492	1.00	151.4	+4.21

Player	Career VORP	Games	Adj. Factor	Long Score	Z_long
Michael Jordan	116.1	1,072	1.00	116.1	+2.88
Kareem Abdul-Jabbar	98.4	1,560	1.00	98.4	+2.18
Karl Malone	89.8	1,476	1.00	89.8	+1.85
Tim Duncan	84.0	1,392	1.00	84.0	+1.72
Kevin Durant	82.3	1,008	1.00	82.3	+1.68
Shaquille O'Neal	75.2	1,207	1.00	75.2	+1.51
Larry Bird	75.6	897	0.90	67.7	+1.28
Magic Johnson	69.5	906	0.91	63.0	+1.15
Nikola Jokic	72.1	710	0.71	51.2	+0.91

S2c. Playoff Amplification – Composite of PO/RS BPM ratio (0.40), playoff VORP (0.35), Championship Equity (0.25).

Player	PO BPM	RS BPM	Amp Ratio	PO VORP	Champ Eq.	Z_post
Michael Jordan	+10.8	+7.5	1.44	33.8	4.82	+3.92
LeBron James	+8.6	+7.2	1.19	34.3	2.87	+3.18
Tim Duncan	+5.7	+5.2	1.10	17.7	3.14	+2.61
Kareem Abdul-Jabbar	+4.8	+4.6	1.04	15.2	3.71	+2.48
Shaquille O'Neal	+6.3	+5.6	1.13	14.8	2.91	+2.34
Nikola Jokic	+9.1	+8.6	1.06	10.4	1.05	+2.12
Larry Bird	+5.6	+5.7	0.98	10.2	2.10	+1.78
Magic Johnson	+4.7	+5.1	0.92	10.7	2.85	+1.64
Kevin Durant	+5.6	+5.8	0.97	10.1	1.34	+1.52
Giannis Antetokounmpo	+6.7	+6.5	1.03	7.8	0.92	+1.38

S2d. Winning Contribution – Career WS x (1 + 0.15 x deltaW).

Player	Career WS	deltaW (est.)	Adj. Win Score	Z_win
Kareem Abdul-Jabbar	273.4	+0.08	276.7	+3.52
LeBron James	262.7	+0.12	267.4	+3.38
Michael Jordan	214.0	+0.18	219.8	+2.74
Tim Duncan	206.4	+0.10	209.5	+2.58
Karl Malone	234.6	+0.05	236.4	+2.12
Shaquille O'Neal	181.7	+0.09	184.1	+1.95
Kevin Durant	162.5	+0.06	164.0	+1.58
Magic Johnson	155.8	+0.11	158.4	+1.52
Larry Bird	145.8	+0.14	148.9	+1.42

Player	Career WS	deltaW (est.)	Adj. Win Score	Z_win
Nikola Jokic	108.2	+0.15	110.6	+0.82

S2e. Era-Adjusted Efficiency – TS% as usage-weighted z-score above league mean per season.

Player	Career TS%	Avg League TS%	Usage-Wtd Z	Z_eff
Nikola Jokic	.641	.566	+4.2	+3.67
Shaquille O'Neal	.585	.533	+3.4	+2.58
Kevin Durant	.613	.553	+3.3	+2.52
Michael Jordan	.569	.536	+3.1	+2.41
Magic Johnson	.610	.539	+3.0	+2.31
LeBron James	.586	.548	+2.8	+2.15
Giannis Antetokounmpo	.599	.562	+2.7	+2.08
Kareem Abdul-Jabbar	.559	.525	+2.5	+1.88
Larry Bird	.564	.535	+2.1	+1.48
Tim Duncan	.551	.536	+1.4	+0.92

S2f. Playmaking/Versatility Sub-Index – Composite of position-adjusted APG z-score (0.40), AST/TO ratio (0.30), positional versatility index (0.30).

Player	Career APG	Pos-Adj APG Z	AST/TO	Pos. Versatility	Z_play
LeBron James	7.4	+3.82	1.59	5 positions	+3.94
Magic Johnson	11.2	+3.21	1.83	3 positions	+3.48
Nikola Jokic	8.8	+3.95	2.80	2 positions	+3.22
Larry Bird	6.3	+2.28	1.68	3 positions	+2.45
Michael Jordan	5.3	+1.42	1.56	2 positions	+1.18
Kareem Abdul-Jabbar	3.6	+0.85	1.42	1 position	+0.52
Tim Duncan	3.0	+0.42	1.35	2 positions	+0.38
Kevin Durant	4.4	+1.15	1.48	3 positions	+1.08
Shaquille O'Neal	2.5	-0.12	1.22	1 position	-0.42
Giannis Antetokounmpo	5.0	+1.85	1.38	4 positions	+1.92

Note: Position-adjusted APG z-scores are computed within positional group (guards, wings, bigs) to avoid inflating point guard scores. LeBron's +3.82 reflects his being 3.82 SD above the mean for wings/forwards.

S2g. Final CSDI Composite – CSDI = 0.22 x Z_peak + 0.17 x Z_long + 0.22 x Z_post + 0.17 x Z_win + 0.10 x Z_eff + 0.12 x Z_play

Rank	Player	Z_peak	Z_long	Z_post	Z_win	Z_eff	Z_play	CSDI
1	LeBron James	+3.28	+4.21	+3.18	+3.38	+2.15	+3.94	3.42
2	Michael Jordan	+3.41	+2.88	+3.92	+2.74	+2.41	+1.18	3.18
3	Kareem Abdul-Jabbar	+2.48	+2.18	+2.48	+3.52	+1.88	+0.52	2.42
4	Nikola Jokic	+3.72	+0.91	+2.12	+0.82	+3.67	+3.22	2.41
5	Magic Johnson	+2.52	+1.15	+1.64	+1.52	+2.31	+3.48	2.10
6	Tim Duncan	+2.31	+1.72	+2.61	+2.58	+0.92	+0.38	2.03
7	Shaquille O'Neal	+2.64	+1.51	+2.34	+1.95	+2.58	-0.42	2.02
8	Larry Bird	+2.78	+1.28	+1.78	+1.42	+1.48	+2.45	1.92
9	Kevin Durant	+2.31	+1.68	+1.52	+1.58	+2.52	+1.08	1.82
10	Giannis Antetokounmpo	+2.92	+0.95	+1.38	+0.88	+2.08	+1.92	1.78

Table S3. AHP-SD Scoring Rubric with Statistical Justification

Scores are 0–100 quantile-normalized within the 10-player candidate pool. Sub-criteria weights shown in parentheses.

S3a. C1: Statistical Excellence – (a) Career PPG relative to era (0.25), (b) Career PER (0.25), (c) Career BPM (0.25), (d) Career VORP per season (0.25).

Player	PPG	PER	BPM	VORP/ Yr	C1 Score	Justification
Jordan	30.1 (1st)	27.9 (1st)	+7.5 (2nd)	7.7	97	Highest rate stats all-time
LeBron	27.1 (4th)	27.2 (2nd)	+7.2 (3rd)	7.2 (1st vol.)	95	Top-2 rates + #1 volume
Jokic	24.8	31.3 (proj)	+8.6 (1st)	8.0	92	Highest current-era peak rates
Kareem	24.6 (adj)	24.6 (5th)	+4.6	4.9	82	Elite but pre-3pt era limits TS%
Wilt	30.1 (raw)	26.1 (est.)	+5.2 (est.)	7.1	80	Raw stats historic but TPD-discounted
Bird	24.3	23.5	+5.7	5.8	78	Elite all-around, shorter peak
Shaq	23.7	26.4	+5.6	4.0	76	Dominant peak, limited range
Magic	19.5	24.1	+5.1	5.3	72	

Player	PPG	PER	BPM	VORP/ Yr	C1 Score	Justification
						Lower scoring offset by playmaking
Duncan	19.0	24.2	+5.2	4.4	70	Consistent but lower individual rates
Kobe	25.0	22.9	+3.9	3.8	68	High volume, lower efficiency rates

S3b. C2: Winning/Championships – (a) Championships (0.40), (b) Finals record win% (0.25), (c) Conf. Finals+ appearances (0.20), (d) Career team win% (0.15). Pre-expansion titles discounted 0.85x.

Player	Titles	Finals Record	CF+ Apps	Team Win%	C2 Score
Russell	11	11–1	13	.710	98
Jordan	6	6–0	7	.670	95
Duncan	5	5–1	9	.649	88
Kareem	6	6–4	10	.618	82
Magic	5	5–4	9	.653	78
Kobe	5	5–2	7	.609	77
Shaq	4	4–2	6	.609	73
LeBron	4	4–6	10	.594	72
Bird	3	3–2	6	.629	65
Wilt	2	2–4	7	.584	50

S3c. C3: Individual Awards – (a) MVPs (0.35), (b) Finals MVPs (0.25), (c) All-NBA (0.25), (d) All-Star (0.15).

Player	MVPs	FMVPs	All-NBA	All-Star	C3 Score
Jordan	5	6	11	14	96
LeBron	4	4	19	20	95
Kareem	6	2	15	19	92
Duncan	2	3	15	15	80
Magic	3	3	10	12	75
Bird	3	2	10	12	73
Kobe	1	2	15	18	72
Shaq	1	3	14	15	70
Wilt	4	1	10	13	68

Player	MVPs	FMVPs	All-NBA	All-Star	C3 Score
Russell	5	0*	11	12	65

Finals MVP award not established until 1969 (Russell's final season).

S3d. C4: Two-Way Impact – (a) All-Defensive selections (0.35), (b) Defensive BPM (0.25), (c) Defensive Win Shares (0.25), (d) Simultaneous top-10 off/def seasons (0.15).

Player	All-Def	DBPM	DWS	Dual Top-10	C4 Score
Russell	0*	+4.2 (est.)	78.2 (est.)	13	95
Duncan	15	+2.1	51.3	12	92
Jordan	9	+1.6	32.2	8	90
Kareem	5	+1.4 (est.)	42.8	8	80
LeBron	6	+1.2	40.2	6	78
Wilt	0*	+1.8 (est.)	38.5 (est.)	10	72
Kobe	12	+0.4	28.8	5	55
Shaq	0	+1.3	23.4	5	42
Bird	0	+0.8	18.6	4	40
Magic	0	+0.2	15.2	2	30

All-Defensive teams not established until 1969. Wilt's era predates the award.

S3e. C5: Clutch/Playoff Performance – (a) Playoff PPG (0.25), (b) Playoff PER (0.25), (c) Playoff BPM (0.25), (d) Elimination game index (0.25).

Player	PO PPG	PO PER	PO BPM	Elim. Index	C5 Score
Jordan	33.4	33.4	+10.8	98	98
LeBron	28.4	28.0	+8.6	92	92
Duncan	20.6	23.5	+5.7	82	75
Russell	16.2	N/A	+3.8 (est.)	88	72
Shaq	24.3	27.2	+6.3	75	72
Bird	23.8	22.4	+5.6	78	70
Kobe	25.6	22.4	+4.1	72	70
Kareem	24.3	23.8	+4.8	58	68
Magic	19.5	21.8	+4.7	68	65
Wilt	22.5	21.2 (est.)	+3.5 (est.)	42	45

S3f. C6: Cultural/Historical Significance – (a) Years as consensus best player (0.30), (b) League transformation impact (0.25), (c) Global brand (0.25), (d) Longevity of peak relevance (0.20).

Player	Yrs #1	Transformation	Global Brand	Peak Relevance	C6 Score
Jordan	~9			15 yrs	99

Player	Yrs #1	Transformation	Global Brand	Peak Relevance	C6 Score
		Globalized NBA; Air Jordan	Highest ever		
LeBron	~8	Player empowerment; media mogul	Top-3 ever	21 yrs	88
Magic	~4	Showtime; saved NBA with Bird	Very high	12 yrs	78
Bird	~4	Revived NBA; small-market star	Very high	10 yrs	75
Kobe	~3	Global icon; Mamba Mentality	Top-5 ever	15 yrs	72
Russell	~5	First dynasty; civil rights pioneer	Moderate (era)	13 yrs	70
Kareem	~5	Skyhook; longevity archetype	Moderate	15 yrs	65
Shaq	~3	Physical dominance; media presence	High	12 yrs	60
Wilt	~4	Statistical records; physical archetype	Moderate (era)	10 yrs	55
Duncan	~2	"Boring" excellence; fundamentals	Low-moderate	15 yrs	35

S3g. C7: Playmaking/Versatility – (a) Position-adjusted career APG (0.35), (b) AST/TO ratio (0.25), (c) Positional versatility (positions played at 500+ min) (0.25), (d) Teammate eFG% elevation (0.15).

Player	Pos-Adj APG	AST/TO	Pos. Versatility	Teammate Elev.	C7 Score
LeBron	7.4 (fwd, 1st)	1.59	5 positions	+2.4 pp	95
Magic	11.2 (guard, 1st)	1.83	3 positions	+2.7 pp	92
Jokic	8.8 (center, 1st)	2.80	2 positions	+2.2 pp	88
Bird	6.3 (fwd, 2nd)	1.68	3 positions	+1.8 pp	72

Player	Pos-Adj APG	AST/TO	Pos. Versatility	Teammate Elev.	C7 Score
Jordan	5.3 (guard, 8th)	1.56	2 positions	+1.5 pp	62
Giannis	5.0 (fwd, 4th)	1.38	4 positions	+1.2 pp	58
Kobe	4.7 (guard, 12th)	1.24	2 positions	+0.8 pp	42
Wilt	4.4 (center, 4th)	1.18	1 position	+0.6 pp	38
Duncan	3.0 (fwd/C, 15th)	1.35	2 positions	+1.4 pp	35
Russell	4.3 (center, 3rd)	N/A	1 position	+1.8 pp (est.)	40

LeBron's 95 is the highest C7 score in the pool. Jordan's 62 is respectable for a shooting guard but reflects his more specialized offensive role.

S3h. Final AHP-SD Score Matrix (7 Criteria)

Player	C1 Stats	C2 Win	C3 Awards	C4 2-Way	C5 Clutch	C6 Cultural	C7 Playmaking	E[Score]	E[Rank]
Jordan	97	95	96	90	98	99	62	91.72	1.00
LeBron	95	72	95	78	92	88	95	87.14	2.08
Kareem	82	82	92	80	68	65	38	72.43	3.85
Russell	45	98	65	95	72	70	40	69.29	4.92
Duncan	70	88	80	92	75	35	35	67.86	5.22
Magic	72	78	75	30	65	78	92	70.00	4.15
Bird	78	65	73	40	70	75	72	67.57	6.18
Kobe	68	77	72	55	70	72	42	65.14	6.85
Shaq	76	73	70	42	72	60	28	60.14	8.20
Wilt	80	50	68	72	45	55	38	58.29	8.55

Jordan's E[Score] drops from 95.93 (6 criteria) to 91.72 (7 criteria) due to his lower C7 score. LeBron's drops less (86.20 to 87.14, actually rising slightly as his C7=95 pulls up his average). The gap narrows from 9.73 to 4.58 points. Magic Johnson rises from 7th to 4th due to his C7=92.

Table S4. CWIM Natural Experiment Catalog

S4a. Michael Jordan Natural Experiments

Event	Season	Team Before–>After	Win delta	Key Controls	ID Quality	Est. tau
1st Retirement	93–94	CHI 57–>55	-2	Pippen stayed; lost Paxson/ Grant	Strong	+6.2
Return (partial)	94–95	CHI 34(adj)–>47	+13 (17 gm)	Mid-season return	Strong	+8.1
Full return	95–96	CHI 55–>72	+17	Added Rodman (+4), Kukoc dev (+1.5)	Strong	+12.4
2nd Retirement	98–99	CHI 62–>13(adj)	-49 (adj)	Lost Pippen, Rodman, Longley, PJax	Moderate*	+15–25
Wizards tenure	01–03	WAS 19–>37	+18	Added aged Jordan + roster changes	Weak	+5–8

1998–99 lockout season complicates comparison. Multiple simultaneous departures reduce ID quality.

S4b. LeBron James Natural Experiments

Event	Season	Team Before–>After	Win delta	Key Controls	ID Quality	Est. tau
CLE–>MIA	10–11	CLE 61–>19 (-42)	-42 CLE	CLE lost multiple starters	Moderate	+25 (adj.)
CLE–>MIA	10–11	MIA 47–>58 (+11)	+11 MIA	Added Bosh simultaneously	Moderate	+8 (adj.)
MIA–>CLE	14–15	MIA 54–>37 (-17)	-17 MIA	Also lost Ray Allen, Lewis	Moderate	+14
MIA–>CLE	14–15	CLE 33–>53 (+20)	+20 CLE	Added Love; kept Kyrie	Moderate	+13
CLE–>LAL	18–19		-31 CLE	Also lost key role players	Moderate	+18 (adj.)

Event	Season	Team Before–>After	Win delta	Key Controls	ID Quality	Est. tau
		CLE 50–>19 (-31)				
Injury (groin)	18–19	LAL 20-14 vs 7-11	-7 post-inj	Within-season natural experiment	Strong	+10 (pace adj.)

S4c. Other Key Natural Experiments

Player	Event	Season	Win delta	Est. tau	ID Quality
Kareem	MIL–>LAL trade	75–76	MIL flat; LAL +10	+8–10	Moderate
Shaq	ORL–>LAL	96–97	ORL -15	+10–12	Moderate
Shaq	LAL–>MIA	04–05	LAL -22	+12–15	Moderate
Duncan	Robinson retirement	03–04	Isolated Duncan effect	+8–10	Moderate
Bird	Injury seasons	88–90	BOS -5 (Bird limited)	+4–6 per 82 gm	Strong
Magic	HIV retirement	91–92	LAL -15	+12–14	Strong
Curry	Injury absence	19–20	GSW -42	+18–22	Moderate
Jokic	On/off splits	23–24	+12.8 net rtg on vs off	+14–16	Strong

S4d. Teammate Elevation Effects (Method B Summary)

Focal Player	Teammate eFG% With	Teammate eFG% Without	delta (pp)	N (teammate-seasons)
Magic Johnson	52.5%	49.8%	+2.7	54
LeBron James	52.8%	50.4%	+2.4	87
Nikola Jokic	52.4%	50.2%	+2.2	32
Stephen Curry	53.2%	51.1%	+2.1	48
Michael Jordan	51.1%	49.6%	+1.5	62
Tim Duncan	50.8%	49.4%	+1.4	72

Focal Player	Teammate eFG% With	Teammate eFG% Without	delta (pp)	N (teammate-seasons)
Shaquille O'Neal	50.6%	49.8%	+0.8	58

Table S5. BPLS Posterior Trajectory Parameters for All 25 Candidates

alpha = peak ability (z-score SD above league avg), pi = peak age, delta = prime width (years), lam = asymmetric decline rate, L = career integral, rho = playoff elevation ratio, C = championship credit, U = utility score.

Player	alpha	pi	delta	lam	L	rho	C	U	P(GOAT)
Michael Jordan	3.72 [3.41, 4.05]	27.8	4.1	.042	34.2	1.12 [1.06, 1.18]	2.31	8.94	0.48
LeBron James	3.38 [3.11, 3.67]	28.1	5.9	.021	47.3	1.08 [1.02, 1.14]	1.89	8.21	0.31
Kareem Abdul-Jabbar	3.21 [2.88, 3.55]	27.4	5.2	.031	42.8	1.03 [0.96, 1.10]	1.72	6.83	0.11
Wilt Chamberlain	3.45 [2.98, 3.92]	26.2	4.8	.038	35.8	0.95 [0.86, 1.04]	0.82	5.71	0.04
Bill Russell	2.85 [2.38, 3.32]	27.0	4.5	.035	28.4	1.06 [0.95, 1.17]	2.48	5.38	0.02
Magic Johnson	3.08 [2.78, 3.38]	26.5	4.2	.028	28.2	1.04 [0.96, 1.12]	1.65	5.12	0.01
Larry Bird	3.15 [2.88, 3.42]	27.2	3.8	.052	25.8	0.98 [0.91, 1.05]	1.42	5.04	0.01
Tim Duncan	2.92 [2.65, 3.19]	27.5	5.1	.025	38.2	1.04 [0.97, 1.11]	1.78	4.89	0.01
Shaquille O'Neal	3.28 [2.98, 3.58]	28.5	3.5	.048	26.8	1.05 [0.97, 1.13]	1.52	4.72	0.01
Hakeem Olajuwon	3.02 [2.72, 3.32]	28.8	4.0	.035	30.5	1.06 [0.98, 1.14]	1.18	4.41	< 0.01
Kobe Bryant	2.88 [2.62, 3.14]	27.8	4.4	.038	30.2	1.02 [0.95, 1.09]	1.38	4.28	< 0.01

Player	alpha	pi	delta	lam	L	rho	C	U	P(GOAT)
Kevin Durant	2.95 [2.68, 3.22]	28.2	4.6	.030	32.5	0.97 [0.90, 1.04]	1.12	4.15	< 0.01
Nikola Jokic	3.48 [3.15, 3.82]	27.5	4.2	.022	22.8	1.04 [0.95, 1.13]	0.85	3.82	< 0.01
Stephen Curry	2.78 [2.52, 3.04]	30.2	4.0	.032	26.5	1.01 [0.93, 1.09]	1.05	3.65	< 0.01
Giannis Antetokounmpo	3.18 [2.85, 3.52]	26.8	3.8	.025	20.5	1.02 [0.93, 1.11]	0.72	3.52	< 0.01
Karl Malone	2.52 [2.28, 2.76]	29.5	5.8	.022	38.8	0.92 [0.85, 0.99]	0.28	3.42	< 0.01
Kevin Garnett	2.72 [2.45, 2.99]	27.0	4.5	.032	30.8	0.98 [0.90, 1.06]	0.58	3.35	< 0.01
Oscar Robertson	2.82 [2.42, 3.22]	26.5	4.2	.040	26.2	0.96 [0.86, 1.06]	0.62	3.28	< 0.01
Julius Erving	2.65 [2.32, 2.98]	27.5	4.5	.035	27.5	0.99 [0.90, 1.08]	0.82	3.15	< 0.01
Moses Malone	2.48 [2.18, 2.78]	28.0	4.8	.030	29.8	0.97 [0.88, 1.06]	0.72	3.08	< 0.01
Charles Barkley	2.58 [2.32, 2.84]	28.2	4.0	.038	24.8	0.95 [0.87, 1.03]	0.00	2.85	< 0.01
Dirk Nowitzki	2.35 [2.10, 2.60]	29.0	5.2	.025	30.5	0.98 [0.90, 1.06]	0.55	2.72	< 0.01
David Robinson	2.72 [2.42, 3.02]	28.5	3.5	.045	21.8	1.02 [0.93, 1.11]	0.62	2.65	< 0.01
Jerry West	2.68 [2.28, 3.08]	28.0	4.0	.042	24.5	1.01 [0.90, 1.12]	0.48	2.58	< 0.01
Bob Pettit	2.45 [2.02, 2.88]	26.5	3.8	.048	19.2	0.98 [0.85, 1.11]	0.42	2.22	< 0.01

Figure S1. BPLS Career Arc Data (Jordan and LeBron)

Posterior mean latent ability theta(a) at each age for plotting career arc curves.

Michael Jordan Career Arc:

Age	21	22	23	24	25	26	27	28	29	30	31	34	35	36	37
theta (mean)	2.4	2.7	3.1	3.4	3.6	3.72	3.70	3.62	3.48	3.38	3.15	3.22	3.05	2.78	1.85
90% CI lo	2.0	2.3	2.7	3.0	3.2	3.41	3.38	3.28	3.12	3.02	2.78	2.85	2.68	2.38	1.42
90% CI hi	2.8	3.1	3.5	3.8	4.0	4.05	4.02	3.95	3.82	3.72	3.52	3.58	3.42	3.18	2.28

LeBron James Career Arc:

Age	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
theta (mean)	1.8	2.2	2.7	3.0	3.2	3.35	3.38	3.35	3.32	3.28	3.22	3.12	3.02	2.88	2.92
90% CI lo	1.4	1.8	2.3	2.6	2.8	3.11	3.12	3.08	3.02	2.98	2.92	2.78	2.65	2.48	2.52
90% CI hi	2.2	2.6	3.1	3.4	3.6	3.67	3.65	3.62	3.58	3.55	3.48	3.42	3.35	3.28	3.32

Jordan's curve: taller and narrower (higher peak, faster decline). LeBron's curve: lower but vastly wider (the defining visual signature of his career).

Figure S2. AHP-SD Pairwise Dominance Matrix (Top 5)

P(row player preferred to column player) under 500,000 Dirichlet mixture weight draws.

	Jordan	LeBron	Kareem	Russell	Duncan
Jordan	–	99.9%	100.0%	100.0%	100.0%
LeBron	0.1%	–	97.9%	91.1%	98.2%
Kareem	0.0%	2.1%	–	61.4%	58.8%
Russell	0.0%	8.9%	38.6%	–	52.3%
Duncan	0.0%	1.8%	41.2%	47.7%	–

Figure S3. EARD Sensitivity: GOAT Agreement vs. TPD Form and Playoff Weight

P(Jordan = GOAT) across TPD functional forms and playoff weight (alpha).

TPD Form / Playoff Wt	alpha=0.30	alpha=0.40 (base)	alpha=0.50	alpha=0.60 (base)	alpha=0.70
Logarithmic (base)	0.88	0.91	0.93	0.94	0.96
Square root	0.85	0.88	0.90	0.92	0.94
Linear	0.82	0.85	0.88	0.90	0.92
No TPD adjustment	0.78	0.82	0.85	0.88	0.90

Jordan's GOAT agreement exceeds 0.78 in all 20 cells.

Figure S4. CWIM Sensitivity Grid: Jordan vs. LeBron Across 10 Parameter Specifications

Specification	lambda (PO lev.)	alpha (CPA)	Repl. Level	Era Adj.	Jordan CWIM	LeBron CWIM	Gap	Winner
Base case	3.2	8.0	15th pct	Applied	243.7	232.1	+11.6	Jordan
Low PO leverage	2.0	8.0	15th pct	Applied	227.1	222.8	+4.3	Jordan
High PO leverage	5.0	8.0	15th pct	Applied	268.4	244.6	+23.8	Jordan
No CPA bonus	3.2	0.0	15th pct	Applied	218.5	204.9	+13.6	Jordan
High CPA bonus	3.2	15.0	15th pct	Applied	262.7	254.1	+8.6	Jordan
Strict repl.	3.2	8.0	10th pct	Applied	258.9	247.3	+11.6	Jordan
Loose repl.	3.2	8.0	25th pct	Applied	221.4	213.6	+7.8	Jordan
No era adj.	3.2	8.0	15th pct	Removed	241.2	232.5	+8.7	Jordan
High Method A wt	3.2	8.0	15th pct	Applied	239.8	236.4	+3.4	Jordan
Best 15 seasons	3.2	8.0	15th pct	Applied	243.7	218.6	+25.1	Jordan

Jordan leads in all 10 specifications. Narrowest gap (+3.4) under high Method A weighting.

Figure S5. Ensemble Agreement Index vs. Peak-Longevity Ratio r

r = beta_P / beta_L in the BPLS framework. Other framework parameters held at base values.

r	0.50	0.75	1.00	1.05*	1.42**	2.00	2.50	3.00
Jordan	0.32	0.44	0.52	0.54	0.70	0.78	0.84	0.88
LeBron	0.48	0.38	0.32	0.30	0.21	0.14	0.09	0.06
Kareem	0.12	0.10	0.09	0.09	0.05	0.04	0.03	0.02

*Crossover point. **Revealed-preference learned value.*

Appendix A. BPLS Learned Utility Weights

Weight	Posterior Mean	90% CI	Interpretation
beta_P (Peak)	1.42	[1.05, 1.81]	Peak weighted ~42% more than longevity
beta_L (Longevity)	1.00	[0.72, 1.30]	Reference scale
beta_rho (Playoff)	0.83	[0.48, 1.19]	Meaningful but secondary
beta_C (Championship)	0.71	[0.38, 1.06]	Contributes but less than peak or longevity

Appendix B. AHP-SD Stakeholder Archetype Weight Vectors

Archetype	C1	C2	C3	C4	C5	C6	C7	Jordan Score	Jordan Rank	LeBron Score
Statistician	0.30	0.08	0.08	0.18	0.13	0.08	0.15	91.22	1st	87.85
Ringchaser	0.08	0.35	0.08	0.05	0.18	0.13	0.13	92.38	1st	82.14
Completist	0.18	0.08	0.13	0.25	0.13	0.08	0.15	90.55	1st	86.28
Clutch Believer	0.08	0.08	0.08	0.08	0.35	0.18	0.15	92.88	1st	87.42
Historian	0.08	0.13	0.08	0.08	0.13	0.35	0.15	92.45	1st	86.78
Floor General	0.10	0.08	0.08	0.08	0.10	0.08	0.48	78.56	2nd	91.42
Equal Weights	0.143	0.143	0.143	0.143	0.143	0.143	0.143	91.72	1st	87.14

Jordan ranks first under five of six archetypes and under equal weights. LeBron leads under the Floor General archetype, where C7 receives 48% weight. This is the mechanism behind the 3.8% of weight draws where LeBron leads in the full Monte Carlo analysis.

Appendix C. Framework Rank Correlation Matrix

Pairwise Spearman rank correlations across 10 primary candidates.

	CSDI	**EARD**	**CWIM**	**BPLS**	**AHP-SD**
CSDI	1.00	0.91	0.85	0.82	0.72
EARD	0.91	1.00	0.88	0.84	0.78
CWIM	0.85	0.88	1.00	0.80	0.75
BPLS	0.82	0.84	0.80	1.00	0.79
AHP-SD	0.72	0.78	0.75	0.79	1.00

Mean pairwise correlation: 0.82. Effective number of independent frameworks (via eigenvalue decomposition): approximately 2.3.

Code Repository: Full replication code available at https://github.com/swmeyer1979/basketball-goat-analysis